WEEKEND PRACTICUM · SESSION 1 OF 8

1. Define activation before asking for the number

For working data analysts

Demo: Paras runs the build in Codex. Students choose Claude Code, Codex, or Cursor.
Before class: unzip student-kit, open the student folder, run the README readiness prompt or `python3 verify.py ready`.
First 5 minutes: check file access; pair up if blocked.

Synthetic classroom data only.

DECISION FRAMING

Three teams, three denominators, one rollback decision

Should engineering roll back Tuesday's onboarding release?

Team	Denominator	Question it answers
Sales	Accounts created	What happens among newly created accounts?
Growth	Completed onboarding	What happens among onboarding completers?
Engineering	Event attempts	What happens per attempted event?

“Before anyone quotes a rate, name the denominator.”

Choose a denominator. Write who can change it. No answers shown yet.

WORKED EXAMPLE · CLASSROOM SYNTHETIC

Worked example: 50% versus 68% from the same two segments

Two separate synthetic pairs: do not mix their rates.

Segment	eligible	activated	rate
A	100	80	80%
B	900	600	66.7%
Total	1000	680	68%

68%

Pooled 680/1000

Second pair: 80% and 20%, equal weights. **50%** simple mean ignores segment size.

```
1.0 * SUM(activated) / NULLIF(SUM(eligible),0)
```

One row per period and segment. eligible = completed 7-day follow-up; activated = at least once. SQL runs on local files, no warehouse. No eligible accounts returns blank, not 0%.

DEFINITION REVIEW

Before and after: the same data, two definitions

Definition	Numerator	Denominator	Result
Before	Accounts created	Event attempts	Not supplied
After	activated	eligible with 7-day follow-up	68% on synthetic rows

Annotate both rows with the metric owner, source owner, and SQL reviewer who can change or review them.

model.json

`activation_7d`

SQL

`1.0 * SUM(activated) / NULLIF(SUM(eligible),0)`

Before the first question, tick aloud: source, grain, SQL, metric owner.

PAIR BUILD · ANSWERS HIDDEN UNTIL DEBRIEF

Hands-on build, 12 minutes

1. Copy starter/model.json to work/model.json.

2. Edit the metric (activation_7d) and grain.

3. Run python3 verify.py model.

All live lab files are local. No connectors or paid plugins needed beyond your own harness access. Work in pairs; keep your result for debrief.

PEER CHALLENGE · EXIT ARTIFACT

Peer challenge and exit artifact: one approved definition

Swap model.json with another pair

Find one way the metric could mislead. Write it down; no answers until debrief.

Leave with exactly one approved definition, written by the room.

Field	Entry
Definition	
Denominator	
Grain	
Metric owner	
Who can change it	
Approved by	

Complete together after peer review.

“Now we may ask for the number.”